

Example 8.2

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \bar{z}_t$$

Parameters : $\phi_1, \phi_2, \sigma_{\bar{z}}^2$.

Yule-Walker equations

$$e_h = \phi_1 e_{h-1} + \phi_2 e_{h-2}$$

$$\Rightarrow e_h - \phi_1 e_{h-1} - \phi_2 e_{h-2} = 0$$

We have $\underline{e} = \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$; $\underline{r} = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$ sample autocorrelation
True autocorrelations

$$R_2 = \begin{pmatrix} 1 & r_1 \\ r_1 & 1 \end{pmatrix}$$

Using the Yule-Walker equations

$$\hat{\underline{\phi}} = \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = R_2^{-1} \underline{r}_2$$

$$= \frac{1}{1 - r_1^2} \begin{pmatrix} 1 & -r_1 \\ -r_1 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

$$= \frac{1}{1 - r_1^2} \begin{pmatrix} r_1(1 - r_2) \\ r_2 - r_1^2 \end{pmatrix}$$

$$\hat{\phi}_1 = \frac{r_1(1 - r_2)}{1 - r_1^2} ; \quad \hat{\phi}_2 = \frac{r_2 - r_1^2}{1 - r_1^2}$$

$$\hat{\sigma}_z^2 = c_0 (1 - \hat{\phi}_1 r_1 - \hat{\phi}_2 r_2).$$